Fast Correlation Function Estimation via Nearest-Neighbor Distribution Ladders

T. Abel *et al.*
KIPAC / Stanford University

Draft — March 2026

Abstract

The two-point correlation function $\xi(r)$ is the primary summary statistic for the clustering analysis of galaxy surveys. Its estimation via the Landy–Szalay estimator requires counting all pairs of objects at each separation, with computational cost scaling as $O(N^2)$ or, with tree acceleration, $O(N \cdot N_{\text{shell}})$ per radial bin. We show that the k-Nearest Neighbor Cumulative Distribution Functions (kNN-CDF), combined with a geometric ladder of diluted subsamples, provide an estimator of $\xi(r)$ that is mathematically equivalent to the Landy–Szalay estimator in the limit of fine binning, while requiring only $O(N_R \log N)$ computation. We develop the full formalism including survey-geometry random catalogs and per-object weights (FKP, completeness, systematic), and demonstrate exact equivalence at the two-point level. We analyze the error budget—showing where the kNN approach matches, exceeds, or falls short of brute-force pair counting—and describe the dilution ladder that extends the measurement to scales far beyond the mean inter-particle separation with built-in variance estimation from overlapping rungs. The method requires no choice of radial binning, returns $\xi(r)$ as a continuous function, and provides all scales simultaneously from a single tree construction per dilution level.

Contents

1 Introduction **3**

2 Background: Nearest-Neighbor Distributions and Pair Counts **4**
2.1 Pair counts as integrals of the density field . . . 5
2.2 The kNN-CDF and its connection to pair counts . . . 5
2.3 The key observation . . . 6

3 The kNN Estimator for $\xi(r)$ **6**
3.1 From CDFs to pair counts . . . 6
3.2 The kNN estimator . . . 7
3.3 Equivalence to Landy–Szalay . . . 7
3.4 The $k = 1$ estimator: simplest case . . . 8
3.5 The summed estimator: using all k . . . 8

4 Survey Geometry and Per-Object Weights **9**
4.1 Survey random catalogs . . . 9
4.2 The need for per-object weights . . . 9
4.3 Weighted kNN-CDFs . . . 9

4.4 Practical implementation . . . 10
4.5 Weights on the random catalog . . . 10
4.6 The interaction between weights and the survival correction . . . 11
4.7 Comparison with weighted LS . . . 11
4.8 Summary of the weighted estimator . . . 11

5 The Dilution Ladder **11**
5.1 Construction . . . 12
5.2 Overlap as a consistency check . . . 12
5.3 Variance estimation from the partition . . . 12
5.4 The full ladder . . . 13
5.5 Combining across levels . . . 13

6 Error Budget **13**
6.1 Sources of error . . . 13
6.2 The DR term: independent query points . . . 14
6.3 The DD term: correlated query points . . . 14
6.4 Optimal k for a given scale . . . 14
6.5 When does the kNN approach win? . . . 15
6.6 Where the kNN approach requires care . . . 15
6.7 Covariance of the $\xi(r)$ measurement . . . 15

7 Non-Poissonian Diagnostics and Covariance from kNN Residuals **16**
7.1 The Poisson$|\xi$ prediction . . . 16
7.2 The non-Poissonian residual . . . 17
7.2.1 Source (i): non-Gaussian density field . . . 17
7.2.2 Source (ii): non-Poisson point process . . . 17
7.3 From Δ_k to the counts-in-cells excess . . . 17
7.4 Connection to the factorial cumulants . . . 18
7.5 The excess variance and its connection to $\alpha_{\rm SN}$ . . . 19
7.6 Separation of density non-Gaussianity from point-process effects . . . 20
7.7 Explicit recipe for improved covariance estimation . . . 20
7.8 Super-sample covariance from dilution subsamples . . . 21
7.9 Diagnostics for observational systematics . . . 21
7.10 Gaussianity test from higher-k CDFs . . . 21

8 Anisotropic Measurements: Multipoles and Projected Statistics **21**
8.1 Legendre multipoles from value-weighted pair counts . . . 22
8.2 The anisotropic metric for projected statistics . . . 22
8.3 Ellipsoidal generating function . . . 23
8.4 What λ provides . . . 23
8.5 Practical workflow . . . 24

9 Implementation: Tree Reuse and Spatial Decomposition **24**
9.1 In-place KD-trees and the implicit octant hierarchy . . . 24
9.2 Spatial subsamples vs. random subsamples . . . 25
9.3 Boundary-correct post-stratification . . . 25
9.4 Nested variance hierarchy as a Gaussianity test . . . 25

9.5 Statistical products from the spatial decomposition 26

10 Validation Strategy **27**
10.1 Test 1: Exact recovery of $\xi(r)$ from Poisson mocks 27
10.2 Test 2: Equivalence with Corrfunc on N-body mocks 27
10.3 Test 3: Dilution ladder consistency 27
10.4 Test 4: Survey weights 28
10.5 Test 5: Non-Poissonian residuals 28
10.6 Test 6: Computational scaling 28
10.7 Figure plan 29

11 Discussion and Outlook **29**
11.1 Relation to previous work 29
11.2 Angular correlation functions 29
11.3 Covariance estimation 30
11.4 Computational cost 30
11.5 Summary 31

1 Introduction

The two-point correlation function $\xi(r)$, or equivalently the power spectrum $P(k)$, is the foundation of clustering analysis in cosmology. Every major galaxy redshift survey—from CfA and SDSS through BOSS to the ongoing DESI and upcoming Euclid and Rubin LSST programs—relies on precise measurements of $\xi(r)$ to constrain cosmological parameters, detect Baryon Acoustic Oscillations, and characterize the growth of structure.

The standard estimator for $\xi(r)$ from a galaxy catalog is the Landy–Szalay (LS) estimator [1]:

$$\hat{\xi}_{\rm LS}(r) \;=\; \frac{\mathrm{DD}(r) - 2\,\mathrm{DR}(r) + \mathrm{RR}(r)}{\mathrm{RR}(r)}\,, \tag{1}$$

where $\mathrm{DD}(r)$, $\mathrm{DR}(r)$, and $\mathrm{RR}(r)$ are the normalized counts of data–data, data–random, and random–random pairs in a spherical shell at separation r. The random catalog traces the survey selection function—angular mask, radial selection, completeness variations—and the LS estimator is optimal in the sense of minimum variance for a Poisson process [1].

The computational cost of evaluating Eq. (1) is dominated by pair counting. For N_d data points and N_R random points, the brute-force cost is $O(N_d^2 + N_d N_R + N_R^2)$. Tree-based algorithms (KD-trees, ball trees, octrees) reduce this by pruning distant pairs, but the cost remains $O(N \cdot N_{\rm shell})$ per radial bin, where $N_{\rm shell}$ is the typical number of objects in the shell. At large separations, $N_{\rm shell}$ grows as r^2 and the acceleration is modest. For current and next-generation surveys with $N_d \sim 10^7$–10^8 objects, the pair count is a significant computational bottleneck, particularly at large scales where $\xi(r)$ is small and high precision is needed.

Several approaches have been developed to accelerate correlation function estimation: approximate pair counting [11], GPU-accelerated pair counters [7], and Fourier-space methods that avoid pair counting entirely by estimating $P(k)$ on a grid and transforming. The last approach is fast but introduces pixelization and aliasing, requires careful treatment of the survey window function, and does not directly yield $\xi(r)$ without a Fourier transform.

In this paper, we propose an alternative approach based on k-Nearest Neighbor Cumulative Distribution Functions (kNN-CDF) [4]. Rather than counting all pairs at each separation, we query

the spatial tree for the k nearest neighbors of each point—a single operation that simultaneously provides information at all separations—and construct $\xi(r)$ from the resulting distance distributions. The key idea is that the distribution of nearest-neighbor distances is a direct, well-understood functional of $\xi(r)$, and that the ratio of the data–data to data–random distributions isolates the clustering signal, just as the LS estimator does with pair counts.

The method we describe has the following properties:

- **Cost:** $O(N_R \log N_d)$ for the data–random term, $O(N_d \log N_d)$ for the data–data term. No pair counting.
- **No binning:** the output is a continuous function of r, determined by the empirical CDF of nearest-neighbor distances. There is no choice of radial bin width or bin centers.
- **All scales simultaneously:** a single tree query for $k_{\max}$ neighbors returns information at all separations from the nearest-neighbor distance to the $k_{\max}$-th neighbor distance.
- **Large-scale reach via dilution:** a geometric ladder of diluted subsamples (each diluted by a factor of 8 relative to the previous) extends the measurement to arbitrarily large scales, with overlapping rungs providing built-in consistency checks and variance estimates.
- **Survey weights:** per-object weights (FKP, completeness, systematic) are incorporated naturally via value-weighted kNN-CDFs.
- **Reduced RR cost:** for a uniform periodic box, the RR term is the Erlang distribution (analytic, zero cost). For surveys with non-trivial geometry and per-object weights, RR requires a random-to-random kNN query at cost $O(N_R \log N_R)$, replacing the $O(N_R^2)$ pair count.
- **Anisotropic measurements:** Legendre multipoles $\xi_\ell(s)$ and the projected correlation function $w_p(r_p)$ are obtained from the same tree via value-weighting and a query-time metric parameter, with no additional tree construction.
- **Free diagnostics:** the kNN residuals provide a scale-dependent measurement of non-Poissonian structure and non-Gaussian covariance contributions; the nonlinear density variance $\sigma^2(R)$ is measured as a continuous function of smoothing scale; and the spatial tree decomposition yields density-split clustering at all scales simultaneously.

The paper is organized as follows. In Section 2, we review the connection between nearest-neighbor distributions and pair counts. In Section 3, we construct the kNN estimator for $\xi(r)$ and prove its equivalence to the LS estimator. In Section 4, we incorporate survey geometry and per-object weights. In Section 5, we describe the dilution ladder for multi-scale estimation with overlap. In Section 6, we analyze the error budget. In Section 7, we show that the kNN residuals provide a direct, scale-dependent measurement of non-Gaussian and non-Poissonian covariance contributions. In Section 8, we extend the framework to redshift-space multipoles and projected statistics via an anisotropic query metric. In Section 9, we describe the tree implementation, the spatial decomposition and its connection to density-split clustering and excursion set trajectories. In Section 10, we lay out the validation tests needed for adoption. We discuss extensions and conclude in Section 11.

2 Background: Nearest-Neighbor Distributions and Pair Counts

We begin by establishing the precise mathematical relationship between nearest-neighbor distance distributions and the pair-counting operations that underlie the Landy–Szalay estimator.

2.1 Pair counts as integrals of the density field

Consider N_d data points $\{x_i\}$ distributed in a survey volume $\mathcal{V}$ according to an inhomogeneous Poisson process with rate $\lambda(x) = \bar{n}(x)(1 + \delta(x))$, where $\bar{n}(x)$ is the expected number density (encoding the selection function) and $\delta(x)$ is the density contrast. The two-point correlation function is $\xi(r) = \langle \delta(x)\, \delta(x') \rangle$ for $|x - x'| = r$.

The normalized data–data pair count in a thin shell $(r, r + dr)$ around a randomly chosen data point is

$$\mathrm{DD}(r)\, dr \;=\; \frac{1}{N_d} \sum_{i=1}^{N_d} \bar{n}(x_i)\, (1 + \xi(r))\, 4\pi r^2\, dr \;+\; O(\xi^2)\,, \tag{2}$$

where the sum over i samples the density-weighted positions. The data–random pair count is

$$\mathrm{DR}(r)\, dr \;=\; \frac{1}{N_d} \sum_{i=1}^{N_d} \bar{n}(x_i)\, 4\pi r^2\, dr\,, \tag{3}$$

where the randoms trace $\bar{n}(x)$ by construction. The random–random pair count is

$$\mathrm{RR}(r)\, dr \;=\; \langle \bar{n} \rangle\, 4\pi r^2\, dr\,, \tag{4}$$

the expectation for the selection-function-weighted uniform field. The LS estimator (Eq. 1) combines these to isolate $\xi(r)$.

2.2 The kNN-CDF and its connection to pair counts

The k-th nearest neighbor CDF from a set of query points Q to a set of target points T is defined as

$$\mathrm{CDF}_k^{QT}(r) \;\equiv\; \frac{1}{|Q|} \sum_{q \in Q} \mathbf{1}\big(r_k^{QT}(q) \le r\big)\,, \tag{5}$$

where $r_k^{QT}(q)$ is the distance from query point q to its k-th nearest neighbor in T. Taking Q to be the random catalog and T to be the data gives the DR kNN-CDF; taking both Q and T to be the data (excluding self-pairs) gives the DD kNN-CDF.

The derivative of the CDF with respect to r is the PDF of k-th nearest-neighbor distances:

$$\mathrm{PDF}_k^{QT}(r) \;\equiv\; \frac{d}{dr} \mathrm{CDF}_k^{QT}(r)\,. \tag{6}$$

For the DR case (random-to-data), this PDF is related to pair counts by

$$\mathrm{PDF}_k^{\mathrm{DR}}(r) \;=\; 4\pi r^2\, \bar{n}_{\mathrm{eff}}(r)\, \frac{[\bar{n}_{\mathrm{eff}}(r)\, V(r)]^{k-1}}{(k-1)!}\, \exp\!\big[-\bar{n}_{\mathrm{eff}}(r)\, V(r)\big]\,, \tag{7}$$

where $V(r) = \frac{4}{3}\pi r^3$ and $\bar{n}_{\mathrm{eff}}(r)$ is the effective mean density averaged over the survey geometry at scale r. This is the generalized Erlang distribution. For a uniform survey with constant $\bar{n}$, it reduces to the standard Erlang form used in **(author?)** [4].

For the DD case (data-to-data), the PDF acquires a clustering term:

$$\mathrm{PDF}_k^{\mathrm{DD}}(r) \;=\; 4\pi r^2\, \bar{n}_{\mathrm{eff}}(r)\, (1 + \xi_k^{\mathrm{eff}}(r))\, \frac{[\bar{n}_{\mathrm{eff}}(r)\, V_k^{\mathrm{eff}}(r)]^{k-1}}{(k-1)!}\, \exp\!\big[-\bar{n}_{\mathrm{eff}}(r)\, V_k^{\mathrm{eff}}(r)\big]\,, \tag{8}$$

where $\xi_k^{\mathrm{eff}}(r)$ and $V_k^{\mathrm{eff}}(r)$ encode the clustering evaluated at scale r and the cumulative effect of clustering on the survival probability (the exponential factor), respectively. The precise form of these corrections is developed in Section 3.

2.3 The key observation

The ratio of the DD to DR PDFs at any radius r is

$$\frac{\mathrm{PDF}_k^{\mathrm{DD}}(r)}{\mathrm{PDF}_k^{\mathrm{DR}}(r)} \;=\; (1+\xi_k^{\mathrm{eff}}(r)) \;\times\; \mathcal{S}_k(r)\,, \tag{9}$$

where $\mathcal{S}_k(r)$ is a *survival correction* arising from the different exponential factors in Eqs. (7) and (8). This survival correction accounts for the fact that the probability of not having found a closer neighbor is different in the clustered (DD) and unclustered (DR) cases.

For $k = 1$ and $\xi \ll 1$, the survival correction is close to unity and the ratio directly measures $1+\xi(r)$. For higher k and stronger clustering, the correction is computable from the lower-k CDFs, making the estimator self-calibrating.

This ratio plays exactly the same role as DD/RR in the Peebles–Hauser estimator [9, 10], but without the need to count all pairs—only the nearest neighbors.

3 The kNN Estimator for $\xi(r)$

We now construct an explicit estimator for $\xi(r)$ from the measured kNN-CDFs, and demonstrate its equivalence to the Landy–Szalay estimator.

3.1 From CDFs to pair counts

The pair count in a thin shell $(r, r+dr)$ is, by definition, the number of neighbors at distance r averaged over all query points. This is precisely the PDF of nearest-neighbor distances times the number of query points—but summed over *all* neighbors, not just the k-th.

To recover the total pair count from kNN-CDFs, we use the counts-in-cell decomposition [4]. The probability that a sphere of radius r centered on a query point contains exactly k target points is

$$P_{k|r}^{QT} \;=\; \mathrm{CDF}_k^{QT}(r) - \mathrm{CDF}_{k+1}^{QT}(r)\,, \quad k \geq 1\,, \tag{10}$$

with $P_{0|r}^{QT} = 1 - \mathrm{CDF}_1^{QT}(r)$. The mean number of targets within radius r is then

$$\langle N\rangle^{QT}(r) \;=\; \sum_{k=1}^{\infty} k\, P_{k|r}^{QT} \;=\; \sum_{k=1}^{\infty} \mathrm{CDF}_k^{QT}(r)\,. \tag{11}$$

The pair count in a shell $(r, r+dr)$ is the derivative:

$$\frac{d\langle N\rangle^{QT}}{dr}(r) \;=\; \sum_{k=1}^{\infty} \mathrm{PDF}_k^{QT}(r)\,. \tag{12}$$

In practice, the sum is truncated at $k_{\max}$—the maximum number of neighbors queried. On scales where $\mathrm{CDF}_{k_{\max}}^{QT}(r) \approx 1$ (i.e. almost all query points have found $k_{\max}$ neighbors within r), the truncation error is negligible. On scales where $\mathrm{CDF}_{k_{\max}}^{QT}(r) < 1$, the sum underestimates the true pair count because some query points have more than $k_{\max}$ neighbors. The dilution ladder (Section 5) ensures that at every scale of interest, there exists a dilution level where $k_{\max} = 8$ is sufficient.

3.2 The kNN estimator

Define the effective pair-count densities:

$$\mathcal{D}^{\rm DD}(r) \equiv \frac{d\langle N\rangle^{\rm DD}}{dr}(r) = \sum_{k=1}^{k_{\rm max}} {\rm PDF}_k^{\rm DD}(r)\,, \tag{13}$$

$$\mathcal{D}^{\rm DR}(r) \equiv \frac{d\langle N\rangle^{\rm DR}}{dr}(r) = \sum_{k=1}^{k_{\rm max}} {\rm PDF}_k^{\rm DR}(r)\,. \tag{14}$$

These are the total pair-count densities at separation r, summed over all neighbors up to $k_{\rm max}$, for the DD and DR cases respectively.

In the standard pair-counting framework, the pair count in a shell at radius r around a typical data point is $4\pi r^2\,\bar{n}_{\rm eff}(1+\xi(r))$ for DD and $4\pi r^2\,\bar{n}_{\rm eff}$ for DR (Eqs. 2–3). Therefore:

$$\boxed{1+\hat{\xi}(r) \;=\; \frac{\mathcal{D}^{\rm DD}(r)}{\mathcal{D}^{\rm DR}(r)} \;=\; \frac{\sum_{k=1}^{k_{\rm max}} {\rm PDF}_k^{\rm DD}(r)}{\sum_{k=1}^{k_{\rm max}} {\rm PDF}_k^{\rm DR}(r)}\,.} \tag{15}$$

This is the kNN estimator for $\xi(r)$. It is the ratio of the total DD pair-count density to the total DR pair-count density, both reconstructed from the kNN distance distributions.

3.3 Equivalence to Landy–Szalay

We now show that Eq. (15) is equivalent to the LS estimator in the limit where the RR term is computed analytically.

The LS estimator can be rewritten as

$$\hat{\xi}_{\rm LS}(r) \;=\; \frac{{\rm DD}(r)}{{\rm RR}(r)} - 2\frac{{\rm DR}(r)}{{\rm RR}(r)} + 1\,. \tag{16}$$

(author?) [3] showed that this is equivalent to

$$1+\hat{\xi}_{\rm LS}(r) \;=\; \frac{{\rm DD}(r)\cdot{\rm RR}(r)}{[{\rm DR}(r)]^2}\,. \tag{17}$$

In the regime where the random catalog perfectly traces the selection function and $N_R \to \infty$, the DR term converges to its expectation and ${\rm RR}(r)/{\rm DR}(r) \to {\rm DR}(r)/{\rm DD}_{\rm Poisson}(r)$, so that

$$1+\hat{\xi}_{\rm LS}(r) \;\to\; \frac{{\rm DD}(r)}{{\rm DR}(r)}\,, \tag{18}$$

which is the Davis–Peebles estimator [10], and is identical in form to our Eq. (15).

The LS estimator differs from Davis–Peebles by including the RR term, which corrects for fluctuations in the random catalog. In the kNN framework, this correction is unnecessary when N_R is large: the DR kNN-CDF converges to its expectation as $1/\sqrt{N_R}$, and for $N_R \gg N_d$ the fluctuations in DR are negligible compared to the cosmic variance in DD. Moreover, for a Poisson process with known $\bar{n}$, the DR expectation is the Erlang distribution—available analytically—so no random catalog is needed for the reference term at all.

When the selection function is non-trivial and $\bar{n}(x)$ is not constant, the random catalog encodes the geometry. In this case, the DR kNN-CDF from randoms (drawn from $\bar{n}(x)$) to data correctly captures the expected pair count given the survey window, and the ratio in Eq. (15) remains valid.

The LS correction in kNN language. If one wishes to retain the full LS correction for finite random catalogs, the kNN analog is:

$$1 + \hat{\xi}_{\rm LS}^{k\rm NN}(r) \;=\; \frac{\mathcal{D}^{\rm DD}(r)\cdot\mathcal{D}^{\rm RR}(r)}{[\mathcal{D}^{\rm DR}(r)]^2}\,, \tag{19}$$

where $\mathcal{D}^{\rm RR}(r)$ is the RR pair-count density reconstructed from the RR kNN-CDFs (random-to-random queries). For $N_R \gg N_d$, the cost of the RR query is $O(N_R \log N_R)$, which may exceed the DR cost but is still $O(N_R \log N_R)$ rather than $O(N_R^2)$. In practice, $\mathcal{D}^{\rm RR}(r)$ can be computed analytically for simple geometries, or from a much smaller random catalog than used in standard LS, since it only needs to capture the geometry, not beat down shot noise.

3.4 The $k = 1$ estimator: simplest case

For $k_{\rm max} = 1$, the estimator simplifies to

$$1 + \hat{\xi}(r) \;=\; \frac{{\rm PDF}_1^{\rm DD}(r)}{{\rm PDF}_1^{\rm DR}(r)}\,. \tag{20}$$

This is the ratio of the 1NN PDFs. The 1NN PDF at radius r is the pair density at r times the probability of no closer pair (the survival function):

$$\begin{aligned} {\rm PDF}_1^{\rm DR}(r) &= 4\pi r^2\,\bar{n}_{\rm eff}\,\exp\!\big[-\bar{n}_{\rm eff}\,V(r)\big]\,, && (21)\\ {\rm PDF}_1^{\rm DD}(r) &= 4\pi r^2\,\bar{n}_{\rm eff}\,(1+\xi(r))\,\exp\!\big[-\bar{n}_{\rm eff}\,V(r)\,(1+\bar{\xi}_V(r))\big]\,, && (22)\end{aligned}$$

where $\bar{\xi}_V(r) \equiv (3/r^3)\int_0^r \xi(r')\,r'^2\,dr'$ is the volume-averaged correlation function. The ratio is

$$\frac{{\rm PDF}_1^{\rm DD}(r)}{{\rm PDF}_1^{\rm DR}(r)} \;=\; (1+\xi(r))\,\exp\!\big[-\bar{n}_{\rm eff}\,V(r)\,\bar{\xi}_V(r)\big]\,. \tag{23}$$

The exponential factor is the survival correction: in a clustered field, a data point is more likely to have already found a very close neighbor, depleting the 1NN PDF at larger r.

For $\xi \ll 1$ (large scales), $\bar{\xi}_V \ll 1$ and the correction is negligible: the ratio gives $1 + \xi(r)$ directly. For $\xi \sim 1$ (nonlinear scales), the correction matters. However, it depends only on $\bar{\xi}_V(r)$—the integral of ξ at smaller scales—which is measured from the same data at smaller r. The estimator is therefore self-calibrating: one measures $\xi(r)$ starting from the smallest scales (where the correction is largest but the signal is strongest) and propagates outward, using the already-measured small-scale ξ to correct the survival factor at larger scales.

3.5 The summed estimator: using all k

The summed estimator (Eq. 15) combines information from all $k \le k_{\rm max}$. Its virtue is that the survival corrections partially cancel in the sum. The total DD pair-count density $\mathcal{D}^{\rm DD}(r) = \sum_k {\rm PDF}_k^{\rm DD}(r)$ counts *all* neighbors at distance r, regardless of how many closer neighbors exist. Similarly for $\mathcal{D}^{\rm DR}(r)$. The sum over k restores the equivalence with the full pair count, where each pair at separation r is counted once regardless of the pair ordering.

Formally, for $k_{\rm max} \to \infty$:

$$\sum_{k=1}^{\infty} {\rm PDF}_k^{QT}(r) \;=\; 4\pi r^2\,\bar{n}_{\rm eff}^{QT}(r)\,, \tag{24}$$

where $\bar{n}_{\rm eff}^{QT}(r)$ is the mean density of targets at distance r from a typical query point. The survival corrections cancel exactly in the infinite sum, because every neighbor at distance r is the k-th nearest for exactly one value of k. The ratio $\mathcal{D}^{\rm DD}/\mathcal{D}^{\rm DR}$ then gives $1+\xi(r)$ without any survival correction—identical to the standard pair-count ratio.

For finite $k_{\rm max}$, the cancellation is incomplete: the contribution of pairs that are the $(k_{\rm max}+1)$-th or higher neighbor is missing. But on scales where ${\rm CDF}_{k_{\rm max}}^{QT}(r) \approx 1$ (almost all query points have $\geq k_{\rm max}$ neighbors within r), the missing contribution is negligible. The dilution ladder (Section 5) ensures this condition is met at every scale.

4 Survey Geometry and Per-Object Weights

Real galaxy surveys have non-trivial geometry (angular masks, radial selection functions, fiber collisions) and require per-object weights for optimal estimation. In this section we show how these are incorporated into the kNN estimator.

4.1 Survey random catalogs

The standard practice in galaxy clustering analysis is to generate a random catalog of $N_R \gg N_d$ points drawn from the survey selection function $\bar{n}(x)$. These randoms trace the expected number density at every point in the survey volume, encoding the angular mask, the radial selection, and any position-dependent completeness. In the LS estimator, the DR and RR pair counts use these randoms as reference.

In the kNN framework, the randoms play the same role: the DR kNN-CDF measures distances from survey randoms to data. Because the randoms trace $\bar{n}(x)$, the DR CDF automatically accounts for the survey geometry. No modification to the kNN query is needed—the standard survey random catalog is used directly as the set of query points.

4.2 The need for per-object weights

In practice, each data galaxy carries a weight w_i that corrects for various observational effects. The most common weights are:

- **FKP weight** [2]: $w_{\rm FKP}(x) = 1/(1+\bar{n}(x)\,P_0)$, which optimizes the signal-to-noise ratio for the power spectrum at a fiducial amplitude P_0.
- **Completeness weight**: $w_{\rm comp}(x) = 1/C(x)$, where $C(x)$ is the probability that a galaxy at position x is included in the catalog (e.g. due to fiber assignment).
- **Systematic weight**: corrections for imaging systematics, stellar contamination, etc.

The total weight is typically the product: $w_i = w_{{\rm FKP},i}\, w_{{\rm comp},i}\, w_{{\rm sys},i}$.

In the LS estimator, the weighted pair count is

$$\mathrm{DD}_w(r) \;=\; \frac{\sum_{i\neq j} w_i\, w_j\, \Theta(r_{ij} \in \text{shell})}{\left(\sum_i w_i\right)^2}\,, \tag{25}$$

and similarly for DR and RR. The weights modulate each pair's contribution to the total.

4.3 Weighted kNN-CDFs

In the kNN framework, per-object weights are incorporated through the value-weighted kNN-CDF formalism. For each query point q (a random point for DR, a data point for DD), the k nearest

data neighbors are found, and the weighted count is

$$W_k(q) \;\equiv\; \sum_{j=1}^{k} w_{(j)}\,, \tag{26}$$

where $w_{(1)}, \ldots, w_{(k)}$ are the weights of the k nearest neighbors, ordered by distance.

The weighted pair-count density at separation r is then

$$\mathcal{D}_w^{QT}(r) \;\equiv\; \frac{d}{dr}\Big[\frac{1}{|Q|}\sum_{q\in Q} W(q,r)\Big]\,, \tag{27}$$

where $W(q,r) = \sum_{j:r_j\le r} w_{(j)}$ is the cumulative weight within radius r. This is the weighted analog of Eq. (12): instead of counting the number of neighbors at distance r, we sum their weights.

The weighted estimator for $\xi(r)$ is then

$$\boxed{1+\hat{\xi}_w(r) \;=\; \frac{\mathcal{D}_w^{\rm DD}(r)}{\mathcal{D}_w^{\rm DR}(r)}\,.} \tag{28}$$

4.4 Practical implementation

The implementation is straightforward:

1. Build a KD-tree on the data positions $\{x_i\}$.
2. For each query point q (random or data), find the $k_{\rm max}$ nearest data neighbors, returning distances $r_{(1)}, \ldots, r_{(k_{\rm max})}$ and their weights $w_{(1)}, \ldots, w_{(k_{\rm max})}$.
3. Compute the cumulative weight profile $W(q,r)$ as a step function with jumps of $w_{(j)}$ at each $r_{(j)}$.
4. Average $W(q,r)$ over all query points and differentiate to get $\mathcal{D}_w(r)$.
5. Form the ratio $\mathcal{D}_w^{\rm DD}/\mathcal{D}_w^{\rm DR}$.

Step 2 is the only computationally expensive operation, with cost $O(\log N_d)$ per query point. The weights are retrieved from a lookup table indexed by the neighbor's identity (returned by the tree query along with the distance). No additional tree construction or querying is needed beyond the standard kNN search.

4.5 Weights on the random catalog

In some analyses, the random points also carry weights—for example, when the random catalog is generated with a non-uniform density that does not perfectly match the selection function, and a weight is applied to correct the residual mismatch. In the kNN framework, random-point weights enter as weights on the query points themselves:

$$\mathcal{D}_w^{\rm DR}(r) \;=\; \frac{d}{dr}\left[\frac{\sum_{q\in R} w_q^R\, W(q,r)}{\sum_{q\in R} w_q^R}\right]\,, \tag{29}$$

where w_q^R is the weight of random point q. This is a weighted average of the cumulative weight profiles, with the random weights determining the relative contribution of each query point.

4.6 The interaction between weights and the survival correction

For the individual kNN PDFs, the per-object weights interact with the survival correction: a high-weight neighbor at small distance contributes more to the weighted count, potentially affecting the effective survival probability at larger distances. However, in the summed estimator (Eq. 28), the survival corrections cancel just as they do in the unweighted case (Section 3.5). The sum $\mathcal{D}_w(r) = \sum_k \mathrm{PDF}^{QT}_{w,k}(r)$ counts each neighbor once, weighted by $w_{(j)}$, regardless of its rank. The rank ordering is merely a bookkeeping device used by the tree query; the final sum over k erases it.

Therefore, the weighted estimator (Eq. 28) is exact whenever $k_{\max}$ is large enough that virtually all query points have $\geq k_{\max}$ neighbors within radius r. No correction for the interaction between weights and survival is needed.

4.7 Comparison with weighted LS

The weighted kNN estimator (Eq. 28) and the weighted LS estimator (Eq. 25) use the same information—the weighted pair counts—but access it differently. LS sums $w_i\, w_j$ over all pairs at separation r; the kNN approach sums $w_{(j)}$ over nearest neighbors of each query point and then differentiates.

For the DD term, the LS weight is a *product* $w_i\, w_j$, while the kNN weight is a *sum* $\sum w_{(j)}$. These are not identical. The LS product weighting up-weights pairs where *both* members are high-weight; the kNN sum weighting up-weights neighbors that are individually high-weight, regardless of the weight of the query point.

To recover the exact LS product weighting, the kNN query point must also carry a weight. For the DD case, the query point is a data point with weight w_q, and the weighted pair-count density becomes

$$\mathcal{D}^{\mathrm{DD}}_{w\times w}(r) \;=\; \frac{d}{dr}\left[\frac{\sum_{q\in D} w_q\, W(q,r)}{\sum_{q\in D} w_q}\right], \tag{30}$$

which is the weighted average of the cumulative neighbor-weight profiles, with each data point's profile weighted by its own weight. This gives each pair the product weight $w_q \cdot w_{(j)}$, recovering the LS convention exactly.

The DR term is analogous, with random query-point weights w^R_q as in Eq. (29).

4.8 Summary of the weighted estimator

The full weighted kNN estimator, with product weighting matching the LS convention, is:

$$1 + \hat{\xi}_{w\times w}(r) \;=\; \frac{\mathcal{D}^{\mathrm{DD}}_{w\times w}(r)}{\mathcal{D}^{\mathrm{DR}}_{w\times w}(r)}\,. \tag{31}$$

Each term is computed from weighted sums over kNN query results. The weights on query points (data or random) and on their neighbors (data) enter multiplicatively, reproducing the LS pair-weighting scheme. The computational cost remains $O(N_R \log N_d)$ for DR and $O(N_d \log N_d)$ for DD, independent of the weights.

5 The Dilution Ladder

The kNN estimator with $k_{\max}$ neighbors reaches scales up to the $k_{\max}$-th nearest-neighbor distance, which for the undiluted sample is a few times the mean inter-particle separation $\bar{d} = \bar{n}^{-1/3}$. To

extend the measurement to larger scales—where the correlation function is weak but cosmologically important (BAO, large-scale bias, etc.)—we use a geometric ladder of diluted subsamples.

5.1 Construction

Starting from the full data catalog of N_d points, we partition them into a sequence of volume-filling subsamples. At each dilution level ℓ, the data are randomly divided (without replacement) into R_ℓ disjoint subsets, each containing N_d/R_ℓ points distributed throughout the full survey volume:

$$R_0 = 1, \quad R_1 = 8, \quad R_2 = 64, \quad R_3 = 512, \quad \ldots \quad R_\ell = 8^\ell \,. \tag{32}$$

The dilution factor of 8 is chosen so that the mean inter-particle separation increases by a factor of $8^{1/3} = 2$ at each level. Each subset requires its own tree, built on its N_d/R_ℓ points. Because each subset spans the full survey volume, it can measure $\xi(r)$ at all separations up to the survey scale—the dilution increases the mean separation, bringing larger scales within reach of small k.

The k-th nearest neighbor at dilution R_ℓ probes a characteristic scale

$$r_{\rm char}(k, R_\ell) \;\sim\; \left(\frac{k\,R_\ell}{\bar{n}}\right)^{1/3} . \tag{33}$$

With $k = 1, \ldots, 8$ at each dilution level, the range of probed scales is:

$$r_{\rm char}(1, R_\ell) \;\sim\; R_\ell^{1/3}\,\bar{d}\,, \quad r_{\rm char}(8, R_\ell) \;\sim\; 2\,R_\ell^{1/3}\,\bar{d}\,. \tag{34}$$

Each rung covers a factor of 2 in scale. The factor-of-8 dilution between levels ensures that the $k = 1$ scale at level $\ell + 1$ matches the $k = 8$ scale at level ℓ:

$$r_{\rm char}(1, R_{\ell+1}) = r_{\rm char}(1, 8R_\ell) = 2\,R_\ell^{1/3}\,\bar{d} = r_{\rm char}(8, R_\ell)\,. \tag{35}$$

This produces complete overlap between adjacent rungs.

5.2 Overlap as a consistency check

In the overlap region, two independent measurements of $\xi(r)$ are available:

- From level ℓ, using $k = 8$: the 8th nearest neighbor in a sample of N_d/R_ℓ points.
- From level $\ell + 1$, using $k = 1$: the 1st nearest neighbor in a sample of $N_d/R_{\ell+1} = N_d/(8R_\ell)$ points.

These two measurements use different subsets of the data, with different neighbor ranks and different survival corrections. Agreement between them confirms that:

1. The $k_{\rm max} = 8$ truncation is adequate at level ℓ (no missing pairs).
2. The survival correction (Section 3.4) is handled correctly.
3. The dilution procedure does not introduce biases (e.g. from edge effects or the interaction between dilution and the survey geometry).

Disagreement in the overlap region is a red flag that triggers investigation before the measurement is trusted at larger scales.

5.3 Variance estimation from the partition

At each dilution level R_ℓ, the R_ℓ disjoint subsamples are independent (no shared data points). For each subsample m, the kNN estimator yields $\hat{\xi}_m(r)$. The scatter across subsamples provides an empirical estimate of the variance:

$$\widehat{\rm Var}[\hat{\xi}(r)] \;=\; \frac{1}{R_\ell - 1}\sum_{m=1}^{R_\ell}\left(\hat{\xi}_m(r) - \bar{\xi}(r)\right)^2 , \tag{36}$$

where $\bar{\xi}(r)$ is the mean over subsamples. This is a direct, model-free estimate of the measurement variance, incorporating both Poisson noise and cosmic variance.

The degrees of freedom at each level are naturally well matched to the measurement. At the lowest dilution (level 0, full data), $k = 1, \ldots, 8$ probes only small separations $r \lesssim 2\bar{d}$, where the number of independent spatial volumes in the survey is $V_{\rm survey}/r^3 \gg 1$. No variance estimate from multiple subsamples is needed at this level—the measurement itself has negligible sample variance. At the highest dilution (level L, R_L subsamples), $k = 1, \ldots, 8$ probes scales $r \sim 2^L\bar{d}$, where cosmic variance is important—but there are R_L independent measurements with $R_L - 1$ degrees of freedom. The scale–dilution matching ensures that the variance estimate is well determined everywhere without special treatment.

5.4 The full ladder

The complete measurement procedure is:

1. **Level 0** ($R_0 = 1$): use the full data catalog. Measure DD and DR kNN-CDFs for $k = 1, \ldots, 8$. Compute $\hat{\xi}(r)$ via Eq. (15) from the mean separation $\bar{d}$ to $\sim 2\bar{d}$. No variance estimate at this level (only one realization).
2. **Level 1** ($R_1 = 8$): draw $M_1 \sim 8$ subsamples of $N_d/8$ points. For each, measure DD and DR kNN-CDFs for $k = 1, \ldots, 8$. Compute $\hat{\xi}_m(r)$ for each subsample, covering $\sim 2\bar{d}$ to $\sim 4\bar{d}$. Check overlap with Level 0 at $r \sim 2\bar{d}$. Estimate variance from scatter across subsamples.
3. **Level** ℓ ($R_\ell = 8^\ell$): draw M_ℓ subsamples of N_d/R_ℓ points. Measure, overlap-check, estimate variance. Covers $\sim 2^\ell\bar{d}$ to $\sim 2^{\ell+1}\bar{d}$.
4. Continue until the desired maximum scale is reached.

For a survey with $N_d = 10^7$ and mean separation $\bar{d} \approx 5\,h^{-1}$Mpc, four dilution levels reach $\sim 80\,h^{-1}$Mpc, well past the BAO scale. Five levels reach $\sim 160\,h^{-1}$Mpc.

5.5 Combining across levels

The final $\hat{\xi}(r)$ is assembled from the ladder by selecting, at each r, the measurement from the level and k value whose characteristic scale is closest to r and whose CDF is near 0.5 (optimal sampling; Section 6). In the overlap regions, a weighted average of the two adjacent levels is used, with weights inversely proportional to the estimated variance.

The resulting $\hat{\xi}(r)$ is a continuous function of r, spanning from the smallest inter-particle scale to the largest scale accessible by the survey geometry, with an empirical error bar at every point.

6 Error Budget

We analyze the statistical errors of the kNN estimator and compare to the standard pair-counting approach.

6.1 Sources of error

The error on $\hat{\xi}(r)$ has three contributions:

1. **Shot noise** (Poisson error): the finite number of pairs at separation r contributing to the measurement.
2. **Cosmic (sample) variance**: the finite number of independent large-scale modes in the survey volume.

3. **Estimator noise**: statistical fluctuations in the DR and RR reference terms due to finite random catalogs.

The first two are irreducible—they are set by the data and the survey volume. The third can be made arbitrarily small by increasing N_R.

6.2 The DR term: independent query points

In the kNN framework, the DR term is measured from N_R independent random query points. Each random contributes an independent estimate of the DR pair-count density at every r (via its nearest-neighbor distances). The variance of the DR CDF at any point is therefore

$$\mathrm{Var}[\mathrm{CDF}_k^{\mathrm{DR}}(r)] \;=\; \frac{\mathrm{CDF}_k^{\mathrm{DR}}(r)\,(1-\mathrm{CDF}_k^{\mathrm{DR}}(r))}{N_R}\,, \tag{37}$$

which is the binomial variance of the empirical CDF. This decreases as $1/N_R$ with no saturation—every additional random point provides an independent measurement.

In standard pair counting, the DR variance also scales as $1/N_R$, but the proportionality constant depends on the pair-counting geometry (shell volume, edge corrections). The kNN approach has the same scaling but with a different prefactor that depends on the CDF value. The variance is smallest when $\mathrm{CDF}_k^{\mathrm{DR}}(r) \approx 0.5$ (the peak of the CDF), which is the scale where the k-th neighbor distance is most frequently measured.

6.3 The DD term: correlated query points

In the DD case, the query points are the data points themselves, which are clustered. Two data points that are close together see nearly the same local environment, so their nearest-neighbor distances are correlated. The effective number of independent DD measurements is not N_d but

$$N_{\mathrm{eff}}^{\mathrm{DD}}(r) \;\sim\; \frac{V_{\mathrm{survey}}}{V_{\mathrm{corr}}(r)}\,, \tag{38}$$

where $V_{\mathrm{corr}}(r) \sim r^3$ is the correlation volume at scale r. At small r, $N_{\mathrm{eff}}^{\mathrm{DD}}$ is large (many independent regions) and the DD error is small. At large r, $N_{\mathrm{eff}}^{\mathrm{DD}}$ shrinks and cosmic variance dominates.

This is exactly the same limitation that affects standard pair counting: the variance of $\mathrm{DD}(r)$ at large r is dominated by cosmic variance, not shot noise, and adding more pairs from the same volume does not help. The kNN approach and the standard approach have the same cosmic variance floor.

6.4 Optimal k for a given scale

As discussed in Section 5, the kNN CDF peaks at a characteristic scale $r_{\mathrm{char}}(k) \sim (k/\bar{n})^{1/3}$. At this scale, the CDF is near 0.5 and the binomial variance (Eq. 37) is maximized relative to the CDF value—paradoxically, this is where the *fractional* error on the CDF is smallest, because the denominator is $\mathrm{CDF}(1-\mathrm{CDF})$.

More importantly, the PDF is largest at the peak: $\mathrm{PDF}_k(r)$ is maximized near $r_{\mathrm{char}}(k)$. Since the pair-count density is proportional to the PDF, the signal-to-noise for $\xi(r)$ is maximized when $r \approx r_{\mathrm{char}}(k)$.

For the Poisson case ($\xi = 0$), the optimal k for measuring a small departure $\xi(r_0)$ at scale r_0 is

$$k_{\mathrm{opt}}(r_0) \;\approx\; \frac{4}{3}\pi\,\bar{n}_{\mathrm{eff}}\,r_0^3\,, \tag{39}$$

i.e. the expected number of points in a sphere of radius r_0. At this k, the kNN CDF peaks near r_0, and the measurement has maximum statistical weight.

6.5 When does the kNN approach win?

The kNN approach provides the same information as pair counting—the pair-count density at each separation—but accesses it through a different computational path. The advantages are:

- **Computational cost**: $O(N_R \log N_d)$ for DR instead of $O(N_R \cdot N_{\rm shell})$. The savings are largest at large separations where $N_{\rm shell}$ is large and tree pruning in standard pair counting is least effective.
- **No binning**: the CDFs are continuous; no choice of bin width or bin centers is needed. This avoids the bias from wide bins and the noise from narrow bins.
- **Built-in variance**: the dilution ladder provides empirical variance estimates at every scale, complementing (not replacing) mock-based covariance estimation.
- **Reduced RR cost**: for uniform periodic boxes, the RR term is analytic (Erlang). For weighted surveys, the RR kNN query costs $O(N_R \log N_R)$ rather than $O(N_R^2)$.
- **Free diagnostics**: the non-Poissonian residuals Δ_k, the nonlinear $\sigma^2(R)$, and density-split statistics are byproducts of the same tree queries (Section 7).

The kNN approach does *not* reduce the cosmic variance floor or the Poisson shot noise—these are properties of the data, not the estimator. It provides the same information as pair counting, faster.

6.6 Where the kNN approach requires care

Finite $k_{\rm max}$. The summed estimator (Eq. 15) requires $k_{\rm max}$ large enough that ${\rm CDF}_{k_{\rm max}}(r) \approx 1$ at the scale of interest. If $k_{\rm max}$ is too small, the pair count is underestimated and $\hat{\xi}(r)$ is biased low. The dilution ladder addresses this by ensuring that at every scale, there is a dilution level where $k_{\rm max} = 8$ suffices.

Differentiating the CDF. The pair-count density $\mathcal{D}(r) = d\langle N\rangle/dr$ requires differentiation of the empirical CDF, which is noisy. In practice, the CDF is smoothed (e.g. by kernel regression or spline fitting) before differentiation. The choice of smoothing bandwidth introduces a trade-off between bias and variance, analogous to the choice of bin width in standard pair counting. However, the smoothing is applied to the CDF (a monotone function), which is better conditioned than smoothing a noisy histogram.

Large-scale correlations between DD queries. At large separations, the DD measurements are highly correlated because each data point's local environment overlaps with nearby data points' environments. The effective number of independent measurements is set by the number of independent large-scale modes, not by N_d. The subsample variance from the dilution ladder correctly captures this correlation, but using the full undiluted sample at large r would overstate the precision. The ladder naturally avoids this by using diluted subsamples at large scales, where the subsample points are well separated and their measurements are more nearly independent.

6.7 Covariance of the $\xi(r)$ measurement

An important subtlety: the kNN estimator gives the same $\xi(r)$ as LS, therefore the covariance of the measurement is also the same. The standard covariance decomposition (Gaussian + non-Gaussian + super-sample + non-Poisson shot noise) applies regardless of how $\xi(r)$ was computed.

Any covariance matrix computed for the LS estimator—whether from mocks, the RascalC semi-analytical approach, or analytic models—is directly applicable to the kNN measurement of $\xi(r)$ at the same separations.

This means the kNN approach can be adopted without any change to the downstream inference pipeline: the same covariance matrix, the same likelihood, the same parameter estimation code. The only change is that $\xi(r)$ is computed faster.

However, as we show in Section 7, the kNN framework goes further: the residuals $\Delta_k(r_0)$ between the measured kNN-CDFs and the Poisson$|\xi$ prediction provide a direct, scale-dependent measurement of the non-Gaussian and non-Poissonian contributions that are the hardest to model analytically. These residuals can be used to improve the covariance estimate itself, by replacing the single-parameter shot-noise rescaling in RascalC with a measured, scale-dependent function (Section 7.5).

7 Non-Poissonian Diagnostics and Covariance from kNN Residuals

The kNN-CDFs contain strictly more information than the two-point correlation function. In extracting $\xi(r)$ via the DD/DR ratio (Section 3), we project down to two-point information. The residual—the departure of the measured kNN-CDFs from the prediction of a Poisson process with the measured $\xi(r)$—encodes the non-Poissonian structure of the point process. This residual is available at no additional computational cost and is directly related to the non-Gaussian contributions to the covariance matrix that are the most difficult to model analytically.

7.1 The Poisson$|\xi$ prediction

Given the measured $\hat{\xi}(r)$ and the known mean density $\bar{n}$, the Poisson$|\xi$ model is the hypothesis that the points are drawn from an inhomogeneous Poisson process with rate $\lambda(x) = \bar{n}(1 + \delta(x))$, where δ has the measured two-point statistics and *all higher connected N-point functions vanish.*

Under this hypothesis, the probability generating function for counts in a sphere of volume $V = \frac{4}{3}\pi r_0^3$ is [4]:

$$P^{\mathrm{P}|\xi}(z|V) \;=\; \exp\Big[\bar{n}(z-1)V + \tfrac{1}{2}\bar{n}^2(z-1)^2V^2\sigma_V^2\Big]\,, \tag{40}$$

where

$$\sigma_V^2(r_0) \;=\; \frac{1}{V^2}\int_0^{2r_0} dr\, r^2\, \xi(r)\, W(r|r_0) \tag{41}$$

is the volume-averaged variance, computed from the measured $\hat{\xi}(r)$ via the sphere-overlap weight function $W(r|r_0)$ (Eq. B6 of **(author?)** 4).

From Eq. (40), the Poisson$|\xi$ prediction for the counts-in-cells distribution is:

$$P_{k|V}^{\mathrm{P}|\xi} \;=\; \sum_{m=k}^{\infty} \frac{(-1)^{m-k}}{k!\,(m-k)!}\,\frac{\partial^m}{\partial z^m} P^{\mathrm{P}|\xi}(z|V)\bigg|_{z=0}\,, \tag{42}$$

and the predicted kNN-CDFs follow from $\mathrm{CDF}_{k\mathrm{NN}}^{\mathrm{P}|\xi}(r_0) = \sum_{j=k}^{\infty} P_{j|V}^{\mathrm{P}|\xi}$.

In practice, the Poisson$|\xi$ prediction is computed numerically for each k and r_0 from the measured $\bar{n}$ and $\sigma_V^2(r_0)$, using the standard recurrence relations for the generating function derivatives. The cost is negligible.

7.2 The non-Poissonian residual

Define the residual:

$$\boxed{\Delta_k(r_0) \;\equiv\; \mathrm{CDF}_{k\mathrm{NN}}^{\mathrm{measured}}(r_0) \;-\; \mathrm{CDF}_{k\mathrm{NN}}^{\mathrm{P}|\xi}(r_0)\,.} \tag{43}$$

If the point process is fully characterized by $\bar{n}$ and $\xi(r)$ (Gaussian field, Poisson sampling), then $\Delta_k = 0$ for all k and r_0.

Non-zero Δ_k arises from two physically distinct sources that we now quantify.

7.2.1 Source (i): non-Gaussian density field

Gravitational evolution generates connected N-point functions of the density field at orders $N \geq 3$. These enter the generating function through the cumulant expansion:

$$\log P(z|V) \;=\; \sum_{n=1}^{\infty} \frac{\bar{n}^n (z-1)^n}{n!}\, \bar{\xi}_V^{(n)}\,, \tag{44}$$

where $\bar{\xi}_V^{(n)}$ is the volume-averaged connected n-point function (Eq. A1 of **(author?)** 4). The Poisson$|\xi$ model truncates this at $n = 2$. The $n = 3$ term contributes to Δ_k through the skewness of the density field:

$$\bar{\xi}_V^{(3)} \;=\; \frac{1}{V^3} \int_V\!\!\int_V\!\!\int_V d^3r_1\, d^3r_2\, d^3r_3\; \zeta(r_{12}, r_{23}, r_{31})\,, \tag{45}$$

where ζ is the connected three-point function. The $n = 4$ term involves the volume-averaged connected four-point function (trispectrum), and so on.

7.2.2 Source (ii): non-Poisson point process

Even if the density field were exactly Gaussian, the point process may deviate from Poisson. The key quantity is the *factorial moment* structure of the point process at fixed density. For a Poisson process, the conditional factorial moments are:

$$\langle N(N-1)\cdots(N-j+1)|\lambda\rangle_{\mathrm{Poisson}} \;=\; \lambda^j\,, \qquad \lambda = \bar{n}(1+\delta)\,V\,. \tag{46}$$

Any deviation from this—halo exclusion, satellite multiplicity, fiber collisions—modifies the counts-in-cells distribution and hence the kNN-CDFs. We parameterize the deviation through the *extra variance* at fixed density:

$$\sigma^2_{N|\lambda} \;\equiv\; \mathrm{Var}[N|\lambda] - \lambda \;=\; (\alpha_{\mathrm{local}} - 1)\,\lambda\,, \tag{47}$$

where $\alpha_{\mathrm{local}} = \mathrm{Var}[N|\lambda]/\lambda$ is the local Fano factor. For Poisson, $\alpha_{\mathrm{local}} = 1$. For a sub-Poisson process (exclusion), $\alpha_{\mathrm{local}} < 1$. For a super-Poisson process (intra-halo clustering), $\alpha_{\mathrm{local}} > 1$. Critically, α_{local} may depend on the local density λ and on the sphere radius r_0—it is a function, not a number.

7.3 From Δ_k to the counts-in-cells excess

The non-Poissonian residual in the counts-in-cells distribution follows directly from Δ_k:

$$\delta P_{k|V} \;\equiv\; P_{k|V}^{\mathrm{measured}} - P_{k|V}^{\mathrm{P}|\xi} \;=\; \Delta_k(r_0) - \Delta_{k+1}(r_0)\,, \tag{48}$$

using $P_{k|V} = \mathrm{CDF}_{k\mathrm{NN}}(r_0) - \mathrm{CDF}_{(k+1)\mathrm{NN}}(r_0)$. The $\delta P_{k|V}$ are a redistribution: they sum to zero ($\sum_k \delta P_{k|V} = 0$, since both the measured and predicted distributions are normalized). Their structure reveals the character of the non-Poissonian signal:

- If $\delta P_0 > 0$ and $\delta P_{k>0} < 0$ for low k: more empty spheres than Poisson|ξ predicts—consistent with enhanced void regions from non-Gaussian tails.
- If $\delta P_0 < 0$ and $\delta P_1 > 0$: fewer empty spheres, more singly-occupied ones—consistent with halo exclusion (anti-clustering below the halo scale).
- If $\delta P_{k\gg 1} > 0$: excess of high-multiplicity spheres—consistent with satellite galaxies in massive halos (super-Poisson intra-halo occupation).

7.4 Connection to the factorial cumulants

The factorial cumulants $C_j(V)$ of the counts-in-cells distribution are the central objects in the theory of clustering covariances [11]:

$$C_j(V) \;\equiv\; \bar{n}^j\, \bar{\xi}_V^{(j)} + \text{lower-order shot-noise terms}\,. \tag{49}$$

The first three are:

$$C_1(V) = \bar{n}V\,, \tag{50}$$

$$C_2(V) = \bar{n}V + \bar{n}^2V^2\sigma_V^2\,, \tag{51}$$

$$C_3(V) = \bar{n}V + 3\bar{n}^2V^2\sigma_V^2 + \bar{n}^3V^3\bar{\xi}_V^{(3)}\,. \tag{52}$$

The Poisson|ξ model predicts C_1 and C_2 exactly (by construction), but predicts $C_3 = \bar{n}V + 3\bar{n}^2V^2\sigma_V^2$ (i.e. with $\bar{\xi}_V^{(3)} = 0$). The measured excess is:

$$\delta C_3(V) \;=\; C_3^{\text{measured}} - C_3^{\text{P}|\xi} \;=\; \bar{n}^3V^3\,\bar{\xi}_V^{(3)} + \delta C_3^{\text{non-Poisson}}\,, \tag{53}$$

where $\delta C_3^{\text{non-Poisson}}$ captures any non-Poissonian contribution to the third factorial cumulant.

The kNN residuals Δ_k provide a direct measurement of the factorial moments without binning. We derive this connection explicitly.

The kNN-CDF is the probability that a sphere contains at least k points: $\text{CDF}_{k\text{NN}}(r_0) = P_{>k-1|V} = \sum_{j=k}^{\infty} P_{j|V}$. The falling factorial moments $\langle N^{(j)}\rangle \equiv \langle N(N-1)\cdots(N-j+1)\rangle$ are related to the kNN-CDFs by the identity

$$\langle N^{(j)}\rangle \;=\; j!\sum_{k=j}^{\infty}\binom{k}{j}^{-1}\cdot\binom{k}{j}P_{k|V} \;=\; j!\sum_{k=j}^{\infty}P_{k|V}\;\cdot\;\frac{1}{1}\,, \tag{54}$$

which simplifies to (see Appendix B of **(author?)** 4)

$$\sum_{k=1}^{\infty}\text{CDF}_{k\text{NN}} = \langle N\rangle\,, \qquad \sum_{k=1}^{\infty}(k-1)\,\text{CDF}_{k\text{NN}} = \frac{\langle N^{(2)}\rangle}{2}\,, \tag{55}$$

and more generally $\sum_{k=1}^{\infty}\binom{k-1}{j-1}\text{CDF}_{k\text{NN}} = \langle N^{(j)}\rangle/j!$.

The non-Poissonian excess in the second factorial moment is therefore

$$\delta\langle N^{(2)}\rangle \;=\; 2\sum_{k=1}^{\infty}(k-1)\,\Delta_k(r_0) \;=\; 2\sum_{k=2}^{\infty}(k-1)\,\Delta_k(r_0)\,, \tag{56}$$

since Δ_1 contributes with coefficient $(1-1) = 0$. Note that this is not simply $2\sum_{k\geq 2}\Delta_k$: each Δ_k enters with weight $(k-1)$, giving higher-k residuals proportionally more influence. This weighting

is physically correct—the second factorial moment involves pairs, and the k-th NN CDF counts spheres with $\geq k$ points, each contributing $\binom{k}{2}$ pairs.

For the third factorial moment:

$$\delta\langle N^{(3)}\rangle \;=\; 6\sum_{k=3}^{\infty}\binom{k-1}{2}\Delta_k(r_0) \;=\; 3\sum_{k=3}^{\infty}(k-1)(k-2)\,\Delta_k(r_0)\,. \tag{57}$$

The k-dependent weights $\binom{k-1}{j-1}$ ensure that the j-th factorial moment is most sensitive to the residuals at high k, where the counts-in-cells distribution is probing the tail—exactly where non-Poissonian effects (satellite multiplicity, non-Gaussian tails) produce the largest departures.

From the factorial moments, the excess variance of counts in cells follows:

$$\delta\sigma_N^2(V) \;=\; \delta\langle N^{(2)}\rangle + \delta\langle N\rangle - \delta\langle N\rangle^2 \;=\; \delta\langle N^{(2)}\rangle\,, \tag{58}$$

since $\delta\langle N\rangle = 0$ by construction (the Poisson$|\xi$ model matches the measured mean density). Thus the excess variance is given by Eq. (56), and the connection to the kNN residuals is exact—no approximations beyond the truncation at $k_{\max}$.

7.5 The excess variance and its connection to $\alpha_{\rm SN}$

The total variance of counts in cells is:

$$\sigma_N^2(V) \;=\; \bar{n}V + \bar{n}^2V^2\sigma_V^2 + \delta\sigma_N^2(V)\,, \tag{59}$$

where the first two terms are the Poisson$|\xi$ prediction and $\delta\sigma_N^2(V)$ is the non-Poissonian excess. From Eq. (56) and (58), the excess is directly measured from the residuals:

$$\delta\sigma_N^2(V) \;=\; 2\sum_{k=2}^{\infty}(k-1)\,\Delta_k(r_0)\,. \tag{60}$$

The RascalC shot-noise rescaling parameter [18, 19] is defined as:

$$\alpha_{\rm SN} \;=\; 1 + \frac{\delta\sigma_{N,\rm SN}^2}{\bar{n}V}\,, \tag{61}$$

where $\delta\sigma_{N,\rm SN}^2$ is the shot-noise part of the excess variance. In the RascalC approach, a single $\alpha_{\rm SN}$ is fit to minimize the KL divergence between the semi-analytical and jackknife covariance matrices [19].

The kNN residuals provide a *scale-dependent generalization*:

$$\boxed{\alpha_{\rm SN}(V) \;=\; 1 + \frac{\delta\sigma_N^2(V)}{\bar{n}V}\,,} \tag{62}$$

measured directly from Δ_k at each volume V. This function captures the full scale dependence of the non-Poissonian excess, replacing the single-parameter fit with a direct measurement.

Order of magnitude. For DESI-like LRG samples with $\bar{n} \approx 3\times10^{-4}\,h^3\,{\rm Mpc}^{-3}$, a sphere of radius $r_0 = 10\,h^{-1}\,{\rm Mpc}$ contains $\bar{n}V \approx 1.3$ galaxies on average. At this scale, $\sigma_V^2 \approx 1$, so the Poisson$|\xi$ variance is $\sigma_{N,{\rm P}|\xi}^2 \approx 1.3 + 1.7 \approx 3$. The non-Poissonian correction from the trispectrum and non-Poisson point statistics is $\sim$ 10–30% of this [19], so $\delta\sigma_N^2 \sim$ 0.3–0.9 and $\alpha_{\rm SN} \sim$ 1.2–1.7. This produces $\Delta_k(r_0) \sim$ 0.01–0.03 for the first few k values, well above the statistical noise for $N_R \sim 10^6$ random query points (for which the CDF noise is $\sim 1/\sqrt{N_R} \sim 10^{-3}$). The signal-to-noise ratio for detecting non-Poissonian structure is therefore $\gtrsim 10$ per radial bin—a high-significance measurement from the same data that gave $\xi(r)$.

7.6 Separation of density non-Gaussianity from point-process effects

The excess $\delta\sigma_N^2(V)$ combines two contributions: (i) non-Gaussianity of the density field ($\bar{\xi}_V^{(n)}$ for $n \geq 3$), and (ii) non-Poissonian point statistics. These can be partially separated using the k-dependence and scale dependence of Δ_k.

For a halo-model galaxy population, the separation is clean at the scales of interest. At $r_0 \gg R_{\rm vir}$ (the scales relevant for BAO and full-shape analysis), the intra-halo contribution (non-Poisson) is negligible and $\delta\sigma_N^2$ is dominated by the density trispectrum (non-Gaussian). At $r_0 \lesssim R_{\rm vir}$, the intra-halo satellite distribution dominates, and Δ_k at low k is controlled by the halo occupation distribution (HOD).

The separation can be sharpened by comparing the measured Δ_k to the prediction from the hierarchical ansatz $\bar{\xi}_V^{(n)} = S_n\,(\sigma_V^2)^{n-1}$, where S_n are the hierarchical amplitudes. For the gravitational evolution of an initially Gaussian field, perturbation theory predicts $S_3 \approx 34/7 + \gamma_1$ (where γ_1 is the logarithmic slope of the correlation function), giving a definite prediction for $\delta C_3(V)$. Any residual beyond this is point-process non-Poissonianity, and its scale dependence reveals whether it originates from halo exclusion (deficit at $r_0 \sim 1$–$5\,h^{-1}\,$Mpc), satellite occupation (excess at $r_0 \sim 0.1$–$1\,h^{-1}\,$Mpc), or other effects.

7.7 Explicit recipe for improved covariance estimation

The covariance of the two-point correlation function estimator at separations r_1 and r_2 decomposes as [18]:

$$\mathrm{Cov}[\hat{\xi}(r_1), \hat{\xi}(r_2)] \;=\; C^{\rm G} + C^{\rm NG} + C^{\rm SSC} + C^{\rm SN,\,non\text{-}P}\,, \tag{63}$$

where $C^{\rm G}$ is the Gaussian term (computed by RascalC from ξ and the survey geometry), $C^{\rm NG}$ is the connected trispectrum term, $C^{\rm SSC}$ is the super-sample covariance, and $C^{\rm SN,\,non\text{-}P}$ is the non-Poissonian shot noise.

RascalC approximates $C^{\rm NG} + C^{\rm SN,\,non\text{-}P}$ by rescaling the shot-noise integrals with a single $\alpha_{\rm SN}$—an approximation justified in the squeezed trispectrum limit [18] and validated at the $\lesssim 8\%$ level for BAO error bars [19].

The kNN residuals replace this single-parameter approximation with a direct, scale-dependent measurement. The explicit procedure is:

1. **Measure** $\xi(r)$ from the DD/DR kNN ratio (Section 3).
2. **Compute** $\sigma_V^2(r_0)$ from $\hat{\xi}(r)$ using Eq. (41).
3. **Predict** $\mathrm{CDF}_{k\mathrm{NN}}^{\mathrm{P}|\xi}(r_0)$ from $\bar{n}$ and σ_V^2 using the generating function (Eq. 40).
4. **Compute** $\Delta_k(r_0)$ for $k = 1, \ldots, k_{\rm max}$.
5. **Compute** $\delta\sigma_N^2(V)$ from Eq. (60) and $\alpha_{\rm SN}(V)$ from Eq. (62).
6. **Feed** $\alpha_{\rm SN}(V)$ into RascalC, replacing the constant $\alpha_{\rm SN}$ in the covariance integrals (Eq. 2.6 of **(author?)** 19) with the measured function, evaluated at the scale corresponding to each quadruple's geometry.

Steps 2–5 are analytic or involve trivial numerical operations. Step 6 requires only interpolation of a tabulated function—the computational cost of RascalC is unchanged. This procedure preserves the full RascalC framework (survey geometry, Legendre-binned multipoles, multi-tracer extensions) while replacing its least constrained ingredient—the single-parameter non-Gaussian correction—with a data-driven, scale-dependent measurement.

7.8 Super-sample covariance from dilution subsamples

The super-sample covariance (SSC) arises from density modes on scales larger than the survey volume, which shift the local mean density and cause $\xi(r)$ to fluctuate coherently [20]. No internal estimator can fully capture modes larger than the survey, but the dilution ladder provides a partial handle.

At dilution level R_ℓ, the M_ℓ subsamples see different large-scale environments within the survey. Their scatter:

$$\widehat{\mathrm{Var}}_\ell[\hat{\xi}(r)] \;=\; \frac{1}{M_\ell - 1}\sum_{m=1}^{M_\ell}\big(\hat{\xi}_m(r) - \bar{\xi}_\ell(r)\big)^2 \tag{64}$$

includes shot noise, Gaussian variance, non-Gaussian variance, and SSC from modes within the survey but larger than the subsample separation. Comparing $\widehat{\mathrm{Var}}_\ell$ across dilution levels reveals where the variance stops scaling as $1/M_\ell$—the scale at which super-sample modes dominate. The difference between adjacent levels isolates the SSC contribution from modes between the two dilution scales.

7.9 Diagnostics for observational systematics

Non-Poissonian residuals can also arise from observational effects, each with a characteristic signature in $\Delta_k(r_0)$:

Fiber collisions. In fiber-fed spectrographs, two fibers cannot target galaxies closer than $\sim 62''$ (DESI). This removes close pairs, producing $\Delta_1(r_0) < 0$ at $r_0 \lesssim r_{\mathrm{fiber}}$. The scale and amplitude of this deficit are predicted from the instrument design, providing a consistency check against the applied correction.

Redshift failures and imaging systematics. Density-dependent incompleteness shifts Δ_k at small r_0; angular systematics (stellar density, extinction) produce signatures at large r_0. Computing Δ_k in subregions of different observing quality provides targeted tests.

In all cases, the kNN residuals probe the full counts-in-cells distribution, not just the mean pair count. A systematic that preserves the mean pair density but modifies the variance (e.g. stochastic fiber assignment) is invisible in $\xi(r)$ but detectable in Δ_k.

7.10 Gaussianity test from higher-k CDFs

For a Gaussian density field with Poisson sampling, all kNN-CDFs are determined by $\bar{n}$ and σ_V^2 via Eq. (40). Testing the measured $k = 3, 4, 8$ CDFs against this prediction provides a hierarchy of Gaussianity tests at increasing statistical order—from the same tree query that produced $\xi(r)$. These tests are particularly valuable at the quasi-linear scales ($r_0 \sim 20$–$60\,h^{-1}\,\mathrm{Mpc}$) where the Gaussian approximation to the covariance is least well validated.

8 Anisotropic Measurements: Multipoles and Projected Statistics

Galaxy surveys measure positions in redshift space, where peculiar velocities distort the radial coordinate. The resulting anisotropy in the two-point correlation function—characterized by the Legendre multipoles $\xi_\ell(s)$ or the 2D function $\xi(r_p, \pi)$—is a primary observable for measuring the

growth rate of structure. We show that the kNN framework handles the anisotropic case with minimal modification: the tree and query algorithm are unchanged, and the anisotropy enters through post-processing of each neighbor's separation vector or through a query-time metric parameter.

8.1 Legendre multipoles from value-weighted pair counts

In redshift space, each pair has a separation vector $\mathbf{s}_{qj}$ with magnitude $s = |\mathbf{s}_{qj}|$ and cosine angle to the line of sight $\mu = \mathbf{s}_{qj} \cdot \hat{\mathbf{n}}/s$, where $\hat{\mathbf{n}}$ is the line-of-sight direction (fixed for the plane-parallel approximation, or the midpoint direction for wide-angle corrections).

The Legendre multipoles are:

$$1 + \xi_\ell(s) \;=\; (2\ell+1)\,\frac{\mathcal{D}_\ell^{\mathrm{DD}}(s)}{\mathcal{D}_\ell^{\mathrm{DR}}(s)}\,, \tag{65}$$

where the multipole pair-count density is:

$$\mathcal{D}_\ell^{QT}(s) \;=\; \frac{d}{ds}\left[\frac{1}{|Q|}\sum_{q\in Q}\sum_{j=1}^{k(q,s)}\mathcal{L}_\ell(\mu_{qj})\right]. \tag{66}$$

This is a value-weighted kNN measurement with the weight of each neighbor j set to $v_j = \mathcal{L}_\ell(\mu_{qj})$. The monopole ($\ell = 0$, $v_j = 1$) reduces to the isotropic estimator of Section 3. The quadrupole ($\ell = 2$, $v_j = \frac{1}{2}(3\mu_{qj}^2 - 1)$) and hexadecapole ($\ell = 4$) use the same tree, the same neighbor lists, and the same distance computations; the only additional operation is computing μ_{qj} for each pair (one dot product and one division) and evaluating the Legendre polynomial.

Survival corrections. The tree query finds the k nearest neighbors in Euclidean (redshift-space) distance s. The survival correction—the probability of not finding a closer neighbor—depends on s, not on μ. Since the Legendre weight $\mathcal{L}_\ell(\mu)$ depends on the pair geometry, not on the neighbor rank k, the survival corrections cancel in the sum over k exactly as in the isotropic case (Section 3.5). The summed estimator for multipoles is therefore unbiased.

Wide-angle corrections. For surveys with significant angular extent, the plane-parallel approximation breaks down for wide-angle pairs. In the kNN framework, the wide-angle correction enters only through the definition of μ_{qj}: instead of using a fixed $\hat{z}$, one uses the pair-specific line of sight (e.g. the midpoint direction or the Yamamoto endpoint estimator). The tree query is unchanged.

8.2 The anisotropic metric for projected statistics

For the projected correlation function $w_p(r_p) = 2\int_0^{\pi_{\max}} \xi(r_p, \pi)\, d\pi$ and for the 2D correlation function $\xi(r_p, \pi)$, the relevant neighbor proximity is in the transverse direction r_p, not in Euclidean distance s. A pair with small r_p but large line-of-sight separation π contributes to w_p but would not be among the k nearest Euclidean neighbors.

We address this with a query-time metric rescaling. Define the metric parameter $\lambda \geq 1$ and the weighted distance:

$$d_\lambda^2 \;\equiv\; r_p^2 + \frac{\pi^2}{\lambda^2}\,. \tag{67}$$

For $\lambda = 1$, this is the Euclidean distance s. For $\lambda > 1$, the metric stretches the line-of-sight direction, making it "cheaper" to be far apart along the LOS. In the limit $\lambda \to \infty$, the metric reduces to the transverse distance r_p.

Tree compatibility. The metric rescaling is equivalent to working in coordinates $(x, y, \tilde{z})$ with $\tilde{z} = z/\lambda$. For an in-place KD-tree with round-robin median splits, the tree structure is determined by the rank ordering of coordinates in each dimension. Since rescaling $z \to z/\lambda$ preserves the rank ordering along z (a monotone transformation), the *tree topology is identical for all* λ. The same data permutation, the same median splits, the same subtree structure. Only the distance computation during the query changes: each squared coordinate difference $(\Delta z)^2$ is replaced by $(\Delta z)^2/\lambda^2$ in the distance and pruning calculations.

This means a single tree build serves all λ values. The λ parameter is passed at query time, modifying the distance kernel and the branch-pruning bounds. For BOSQUE, this requires a weighted-norm variant of the query function—a few lines of code in the inner loop, with no change to the tree build or data layout.

8.3 Ellipsoidal generating function

In the λ-metric, the kNN sphere in rescaled coordinates corresponds to a prolate ellipsoid in physical coordinates, with semi-axes $(r, r, \lambda r)$ where r is the k-th neighbor distance in the rescaled metric. The volume is:

$$V_\lambda(r) \;=\; \frac{4}{3}\pi r^3\,\lambda\,. \tag{68}$$

The counts-in-cells generating function generalizes to:

$$P_\lambda(z|V) \;=\; \exp\left[\sum_{n=1}^{\infty} \frac{\bar{n}^n (z-1)^n}{n!}\,\bar{\xi}^{(n)}_{V,\lambda}\right], \tag{69}$$

where $\bar{\xi}^{(n)}_{V,\lambda}$ are the n-point functions volume-averaged over the ellipsoidal region. For $n = 2$:

$$\sigma^2_{V,\lambda}(r_0) \;=\; \frac{1}{V_\lambda^2}\iint_{\text{ellipsoid}} d^3r_1\, d^3r_2\; \xi(|\mathbf{r}_1 - \mathbf{r}_2|)\,, \tag{70}$$

where the integration is over pairs of points within the ellipsoid. The geometric weight function $W_\lambda(r|r_0)$—the overlap volume of two coaxial ellipsoids with semi-axes $(r_0, r_0, \lambda r_0)$ separated by distance r—replaces the sphere-overlap kernel $W(r|r_0)$. For aligned ellipsoids (both oriented along the LOS), W_λ depends on both the magnitude and the LOS-angle of the separation vector.

The full kNN-CDF formalism—the Poisson$|\xi$ prediction, the Δ_k residuals, the factorial cumulants, the $\alpha_{\rm SN}(V)$ extraction—carries over with the substitution $V \to V_\lambda$ and $W \to W_\lambda$. No additional theory is needed.

8.4 What λ provides

The metric parameter λ connects the isotropic and projected statistics in a continuous family:

- $\lambda = 1$: isotropic kNN-CDFs, giving $\xi(s)$ and its multipoles.
- $\lambda = \pi_{\max}/r_{p,\max} \sim 3$–5: kNN-CDFs in an elongated metric optimized for the (r_p, π) measurement. With $k_{\max} = 8$ and appropriate dilution, the k nearest neighbors in this metric efficiently capture pairs with small r_p and large π.
- $\lambda \to \infty$: pure transverse kNN-CDFs, giving $w_p(r_p)$ (up to $\pi_{\max}$ set by the dilution scale).

Additionally, the λ-dependence of the kNN-CDFs and their residuals provides new diagnostic information:

Anisotropic non-Gaussianity. The Δ_k residuals at different λ values probe the anisotropic non-Poissonian structure. Fingers-of-god (virial motions in clusters) produce excess close pairs along the LOS, visible as $\Delta_1 > 0$ at small d_λ for large λ (where the metric emphasizes the LOS direction). Fiber collisions remove close transverse pairs, visible as $\Delta_1 < 0$ at small d_λ for small λ. The λ-dependence separates these effects.

Alcock–Paczyński sensitivity. The AP effect is a geometric distortion that changes the apparent aspect ratio of structures. The λ at which the kNN-CDFs are most consistent with the isotropic Poisson|ξ prediction (minimizing $|\Delta_k|$ summed over k) constrains the AP parameter. This provides an AP measurement from the counts-in-cells distribution, complementary to the standard AP measurement from the ξ multipole ratio.

8.5 Practical workflow

The complete anisotropic measurement proceeds as follows:

1. Build the tree once on the data in physical redshift-space coordinates.
2. Query at $\lambda = 1$ for the Legendre multipoles $\xi_\ell(s)$. For each query–neighbor pair, compute μ_{qj} and accumulate the Legendre-weighted pair-count densities.
3. Query at $\lambda_{\rm proj}$ (chosen to match $\pi_{\rm max}/r_{p,\rm max}$) for the projected statistics. The k nearest neighbors in the $\lambda_{\rm proj}$-metric efficiently capture the relevant pairs for $w_p(r_p)$ and $\xi(r_p, \pi)$.
4. Compute the Δ_k residuals at both λ values. The isotropic residuals probe the spherically-averaged non-Gaussianity; the anisotropic residuals diagnose LOS-specific effects.
5. Optionally, query at intermediate λ values for AP sensitivity or for separating isotropic from LOS-specific non-Gaussianity.

Steps 2–5 use the same tree. Each λ value requires a separate set of queries, with cost $O(N_R \log N_d)$ per λ. The total cost for n_λ metric values is $n_\lambda \times O(N_R \log N_d)$, typically with $n_\lambda = 2$–3.

9 Implementation: Tree Reuse and Spatial Decomposition

The kNN estimator involves two distinct tree-based operations that must be clearly separated. The *dilution ladder* (Section 5) requires volume-filling random subsamples, each with its own tree—these cannot be derived from the full tree. The *spatial decomposition* described in this section uses the structure of the single full-data tree to provide density labels, variance hierarchies, and density-split statistics at no additional cost. These are complementary: the ladder gives $\xi(r)$ and $\mathrm{Var}[\hat{\xi}]$ at all scales; the spatial decomposition gives environmental diagnostics and additional statistical products.

9.1 In-place KD-trees and the implicit octant hierarchy

We use the BOSQUE library [27], which builds KD-trees by in-place partitioning: the data array is permuted so that the tree structure is implicit, with round-robin median splits on the x, y, and z axes. After the build, the left half of the array contains all points below the x-median; within each half, the lower quarter contains points below the y-median; within each quarter, the lower eighth contains points below the z-median.

After 3 levels of splitting, the array is partitioned into $2^3 = 8$ contiguous blocks that tile the survey volume into (approximate) octants. After 3ℓ levels, the array contains 8^ℓ contiguous blocks. Each block is a valid KD-tree subtree: its internal ordering satisfies the implicit tree invariant with the same round-robin convention, starting from the x-axis (since $3\ell \bmod 3 = 0$).

This structure provides the spatial decomposition used for density diagnostics (Section 9.5) and the nested variance hierarchy (Section 9.4). The octant assignment of any point is determined by its position in the sorted array—an $O(1)$ lookup. Note that this hierarchy is a property of the full-data tree and is separate from the dilution ladder, which requires independent, volume-filling trees at each level.

9.2 Spatial subsamples vs. random subsamples

The octant decomposition produces *spatially disjoint* regions. These are fundamentally different from the volume-filling random subsamples used in the dilution ladder.

For $\xi(r)$ estimation: random, volume-filling subsamples are required. Each diluted subset must span the full survey volume so that pairs at all separations are measurable. Each subset needs its own tree. The octant decomposition cannot replace this.

For density diagnostics: the octant decomposition of the full tree provides environmental labels, density-split statistics, and the nested variance hierarchy at zero additional cost. These use the global tree with post-stratification by octant, as described below.

9.3 Boundary-correct post-stratification

For each random query point q in the global DR measurement, we know (from the array index) which octant at each level ℓ it belongs to. The neighbors of q are found from the global tree, correctly including cross-boundary pairs. We then stratify the results:

- Assign each random to its level-ℓ octant m.
- Compute the average of $\hat{\xi}(r)$ over all randoms in octant m, giving the local $\hat{\xi}_m^{(\ell)}(r)$.
- The variance of $\hat{\xi}_m^{(\ell)}(r)$ across the 8^ℓ octants gives the nested variance at level ℓ.

This procedure is exact—no boundary bias—because the neighbors are found globally. The octant labels are metadata, not search constraints.

9.4 Nested variance hierarchy as a Gaussianity test

The nested octant decomposition provides a scale-by-scale decomposition of the total variance of $\hat{\xi}(r)$. At level ℓ, the between-octant variance captures the contribution from density modes at scales between the level-ℓ octant size L_ℓ and the level-$(\ell-1)$ octant size $L_{\ell-1} = 2L_\ell$. The total variance telescopes:

$$\mathrm{Var}_{\mathrm{total}}[\hat{\xi}(r)] \;=\; \sum_{\ell=1}^{L} \mathrm{Var}_{\mathrm{between},\ell-1\to\ell}[\hat{\xi}(r)] \;+\; \mathrm{Var}_{\mathrm{within},L}[\hat{\xi}(r)]\,. \tag{71}$$

For a Gaussian density field, each term is predicted from the power spectrum:

$$\mathrm{Var}^{\mathrm{Gauss}}_{\mathrm{between},\ell-1\to\ell}[\hat{\xi}(r)] \;=\; \frac{2}{V_\ell}\int \frac{d^3k}{(2\pi)^3}\left|W_{\ell-1}(\mathbf{k}) - W_\ell(\mathbf{k})\right|^2 P^2(k)\,, \tag{72}$$

where $W_\ell(\mathbf{k})$ is the window function of the level-ℓ octant. The excess at each level,

$$\delta\mathrm{Var}_\ell \;\equiv\; \mathrm{Var}^{\mathrm{measured}}_{\mathrm{between},\ell-1\to\ell} - \mathrm{Var}^{\mathrm{Gauss}}_{\mathrm{between},\ell-1\to\ell}\,, \tag{73}$$

is the non-Gaussian variance contribution from modes in the scale band $[L_\ell, L_{\ell-1}]$. This provides a scale-resolved non-Gaussianity diagnostic that is independent of the Δ_k residuals (Section 7):

the Δ_k measure non-Gaussianity through the counts-in-cells distribution at each smoothing scale, while the nested variance measures non-Gaussianity through the mode-by-mode variance of $\xi(r)$ itself.

At the coarsest accessible level, the trend of $\delta\mathrm{Var}_\ell$ with ℓ extrapolates to the super-sample covariance contribution from modes larger than the survey.

9.5 Statistical products from the spatial decomposition

The octant hierarchy assigns each galaxy (and each random query point) a local density at each scale R_ℓ, simply from the count in the octant:

$$\hat{\delta}_m^{(\ell)} \;=\; \frac{N_m}{\langle N\rangle_\ell} - 1\,. \tag{74}$$

Combined with the kNN query results, this provides:

Nonlinear $\sigma^2(R)$ on spherical tophats. For each random query point, the kNN distances give the count $N(< R_0)$ within any chosen radius R_0—this is a spherical tophat density estimate. The variance of $N(< R_0)$ across all randoms gives the full nonlinear density variance:

$$\sigma^2_{\mathrm{NL}}(R_0) \;=\; \frac{\mathrm{Var}[N(< R_0)]}{(\bar{n}V_0)^2} - \frac{1}{\bar{n}V_0}\,, \tag{75}$$

where the last term subtracts Poisson shot noise. This is model-free: no Gaussianity assumption, no generating function truncation. It is the full nonlinear answer, including all higher-order contributions. This quantity is available as a continuous function of R_0, at no cost beyond the kNN query.

The derivative $d\ln\sigma^2_{\mathrm{NL}}/d\ln R$ enters the Press–Schechter mass function and its extensions [28, 29]. Because $\sigma^2_{\mathrm{NL}}(R_0)$ is a smooth, densely sampled function of R_0, the derivative is well determined.

Density-split clustering. The density-split clustering (DSC) technique [31, 32, 33] measures $\xi(r)$ separately in subvolumes of different density. The standard approach selects a single smoothing scale and splits into 3–5 density quantiles.

The kNN framework generalizes DSC in three directions simultaneously:

(i) Multi-scale. The nested octant hierarchy provides the density split at every scale R_ℓ simultaneously. Standard DSC at a single scale is a horizontal slice through the excursion set; the nested hierarchy gives the full two-dimensional structure (δ, R).

(ii) Continuous conditioning. The spherical-tophat density $\hat{\delta}(q, R_0)$ from the kNN distances is a continuous variable, not a quantile bin. The conditional correlation function $\xi(r|\delta, R_0)$ can be studied at arbitrary resolution in δ.

(iii) Full kNN-CDF statistics per density bin. Beyond $\xi(r)$ in each density quantile, the kNN-CDFs per quantile probe all N-point functions conditioned on the large-scale density. The Δ_k residuals per quantile test whether the non-Poissonian structure depends on environment—a direct probe of environment-dependent halo occupation.

All of these use the global tree query with post-stratification by density label; no additional tree builds or queries are needed.

Excursion set trajectories. For each galaxy at position x, the nested octant hierarchy gives the smoothed density $\hat{\delta}^{(\ell)}$ at scales R_ℓ, for $\ell = 0, 1, \ldots, L$. The kNN distances from that galaxy's location provide a continuous interpolation of the smoothed density between the discrete octant scales. The sequence $\hat{\delta}^{(\ell)}$ as a function of R_ℓ is the excursion set trajectory for that point—the same object that the extended Press–Schechter formalism [30] uses to predict halo masses, but measured directly from the data.

10 Validation Strategy

We outline the suite of tests needed to validate the kNN estimator for adoption by the LSS measurement community. These tests are organized by the claim they verify, in order of increasing specificity.

10.1 Test 1: Exact recovery of $\xi(r)$ from Poisson mocks

Setup. Generate $N_{\rm mock} = 1000$ Poisson realizations in a periodic box ($L = 2000\,h^{-1}\,\mathrm{Mpc}$, $\bar{n} = 3 \times 10^{-4}\,h^3\,\mathrm{Mpc}^{-3}$, giving $N_d \approx 2.4 \times 10^6$). No clustering: $\xi(r) = 0$.

Test. Compute $\hat{\xi}(r)$ via the kNN estimator (Eq. 15) with $k_{\rm max} = 8$ and $N_R = 10^7$ randoms. Verify:

1. $\langle \hat{\xi}(r) \rangle = 0$ to within statistical precision across all r.
2. The scatter of $\hat{\xi}(r)$ across mocks matches the Poisson prediction: $\mathrm{Var}[\hat{\xi}] = (1+\xi)^2/(N_{\rm pairs}\,\Delta r) \approx 1/(\bar{n}^2 V_{\rm survey}\,4\pi r^2\,\Delta r)$.
3. The kNN-CDFs match the Erlang distribution exactly.
4. $\Delta_k(r_0) = 0$ for all k and r_0 (no non-Poissonian residual).

This verifies the basic machinery and establishes the noise floor.

10.2 Test 2: Equivalence with Corrfunc on N-body mocks

Setup. Use the Quijote simulation suite [26]: 15,000 N-body realizations at fiducial cosmology, each with 512^3 particles in a $(1\,h^{-1}\,\mathrm{Gpc})^3$ box. Construct halo catalogs with FoF or Rockstar, matching DESI LRG number density ($\bar{n} \approx 3 \times 10^{-4}$) via mass thresholding.

Test. For each of 100 randomly selected realizations:

1. Compute $\hat{\xi}_{\rm LS}(r)$ using Corrfunc [7] with 30 linearly spaced bins from 1 to $200\,h^{-1}\,\mathrm{Mpc}$.
2. Compute $\hat{\xi}_{k\rm NN}(r)$ using the kNN estimator at the same separations (evaluating the continuous kNN estimate at the bin centers).
3. Form the difference $\delta\xi(r) = \hat{\xi}_{k\rm NN} - \hat{\xi}_{\rm LS}$ and verify that $\langle \delta\xi \rangle$ is consistent with zero at all r to within $\lesssim 0.1\sigma$.
4. Compare the mock-to-mock scatter of $\hat{\xi}_{k\rm NN}$ and $\hat{\xi}_{\rm LS}$: they must be statistically identical, confirming equivalent information content.

Any systematic bias $\langle \delta\xi \rangle \neq 0$ at any scale would indicate a failure of the estimator and must be traced to its origin (finite $k_{\rm max}$, CDF differentiation, survival correction).

10.3 Test 3: Dilution ladder consistency

Setup. Same Quijote halo catalogs.

Test. At each dilution level $R_\ell = 1, 8, 64, 512$:

1. Compute $\hat{\xi}(r)$ from M_ℓ subsamples using $k = 1, \ldots, 8$.
2. Verify overlap consistency: $\hat{\xi}$ from $k = 8$ at level ℓ agrees with $\hat{\xi}$ from $k = 1$ at level $\ell + 1$ in the overlap region (Section 5.2).
3. Verify that the combined ladder $\hat{\xi}(r)$ agrees with the direct Corrfunc measurement at all scales from 5 to $150\,h^{-1}$ Mpc.
4. Quantify the subsample variance at each level and compare to the mock-to-mock variance (which includes cosmic variance).

10.4 Test 4: Survey weights

Setup. Apply realistic DESI-like weights to the Quijote halo catalogs: FKP weights with $P_0 = 10^4\,h^{-3}\,\mathrm{Mpc}^3$, plus random incompleteness (angular mask with $\sim 70\%$ area coverage, radial selection function $\bar{n}(z)$ declining at high z). Generate a random catalog tracing the selection function.

Test.

1. Compute $\hat{\xi}_w(r)$ with the weighted kNN estimator (Eq. 31) and with weighted Corrfunc.
2. Verify agreement to $\lesssim 0.1\sigma$ at all scales.
3. Check that the product weighting (query weight $\times$ neighbor weight) reproduces the LS convention exactly (Section 4.7).
4. Test sensitivity to the random catalog density: verify convergence as N_R/N_d increases from 10 to 50.

10.5 Test 5: Non-Poissonian residuals

Setup. Same Quijote halo catalogs (non-Poisson due to halo exclusion and the halo mass function).

Test.

1. Measure $\hat{\xi}(r)$ via the kNN estimator.
2. Compute $\sigma_V^2(r_0)$ from $\hat{\xi}$ and predict the Poisson$|\xi$ kNN-CDFs (Eq. 40).
3. Measure $\Delta_k(r_0)$ for $k = 1, \ldots, 8$ across scales $r_0 = 1$–$100\,h^{-1}$ Mpc.
4. Compute $\delta\sigma_N^2(V)$ and $\alpha_{\rm SN}(V)$ (Eqs. 60–62).
5. Compare $\alpha_{\rm SN}(V)$ to the single $\alpha_{\rm SN}$ obtained from the RascalC jackknife pipeline on the same mocks. The scale-averaged $\alpha_{\rm SN}(V)$ should be consistent with the RascalC value.
6. Verify that the Δ_k residuals at $r_0 \lesssim 5\,h^{-1}$ Mpc show the expected halo-exclusion signature ($\Delta_1 < 0$).
7. Feed $\alpha_{\rm SN}(V)$ into RascalC and compare the resulting covariance matrix to the mock sample covariance. The scale-dependent rescaling should reduce the $\lesssim 8\%$ residual found with a single parameter [19].

This is the most demanding test and the one that validates the standout contribution of Section 7.

10.6 Test 6: Computational scaling

Setup. Vary N_d from 10^4 to 10^7 and N_R from 10^5 to 10^8.

Test. Measure wall-clock time for the kNN estimator (tree build + query) and for Corrfunc (pair count) on the same hardware. Verify:

1. The kNN cost scales as $O(N_R \log N_d)$.
2. The crossover point (where kNN becomes faster than Corrfunc) occurs at $N_d \sim 10^5$–10^6.
3. For DESI-scale catalogs ($N_d \sim 10^7$), the speedup is $\gtrsim 100\times$.

10.7 Figure plan

The paper requires the following figures to make the case convincing:

1. **Estimator comparison** (Test 2): $\hat{\xi}_{k\mathrm{NN}}(r)$ vs. $\hat{\xi}_{\mathrm{LS}}(r)$ for a representative Quijote realization, with residuals in a lower panel showing consistency at the $\lesssim 0.1\%$ level.
2. **Dilution ladder** (Test 3): $\xi(r)$ from all dilution levels and k values, color-coded by level, showing overlap consistency and coverage from 5 to $150\,h^{-1}$ Mpc.
3. **Non-Poissonian residuals** (Test 5): $\Delta_k(r_0)$ for $k = 1, 2, 4, 8$ as a function of r_0, showing the halo-exclusion deficit at small r_0 and the non-Gaussian excess at quasi-linear scales. The Poisson$|\xi$ prediction is the zero line.
4. **Scale-dependent** α_{SN} (Test 5): $\alpha_{\mathrm{SN}}(V)$ from the kNN residuals compared to the single RascalC value (horizontal line), showing the scale dependence.
5. **Covariance comparison** (Test 5): correlation matrices from (a) mock sample covariance, (b) RascalC with constant α_{SN}, (c) RascalC with kNN-measured $\alpha_{\mathrm{SN}}(V)$. The kNN-improved version should be closer to the mock truth.
6. **Computational scaling** (Test 6): wall-clock time vs. N_d for kNN and Corrfunc, showing the crossover and the asymptotic $O(N \log N)$ vs. $O(N^2)$ scaling.

11 Discussion and Outlook

11.1 Relation to previous work

The connection between nearest-neighbor distances and the void probability function was recognized by **(author?)** [6]. **(author?)** [4] developed the full kNN-CDF framework and demonstrated its sensitivity to all connected N-point functions. **(author?)** [5] extended it to cross-correlations between two point sets. The present work differs in focus: rather than using kNN-CDFs as a beyond-two-point summary statistic for cosmological parameter estimation, we use them as a *computational shortcut* for the standard two-point correlation function measurement. The goal is not to extract additional information (though the kNN-CDFs do contain higher-order information), but to extract the *same* two-point information faster.

This distinction is important for the intended audience. Survey analysis pipelines are built around $\xi(r)$ and $P(k)$; they have well-characterized covariance models, window function corrections, and systematic error budgets. Introducing a new summary statistic requires re-deriving all of these. By showing that the kNN approach gives the same $\xi(r)$, we allow it to be adopted as a drop-in replacement for the pair-counting step, without changing anything downstream.

11.2 Angular correlation functions

The formalism developed here applies directly to 3D correlation functions measured from spectroscopic surveys. For photometric surveys, the primary observable is the angular correlation function $w(\theta)$, measured from projected positions on the sky. The kNN framework generalizes straightforwardly to 2D: the tree is built on angular coordinates, the CDFs measure angular nearest-neighbor

distances, and the estimator for $w(\theta)$ follows the same DD/DR ratio structure. The computational savings are even larger in 2D, since the pair count grows as $N \cdot N_{\text{annulus}}$ with $N_{\text{annulus}} \propto \theta$ rather than r^2.

11.3 Covariance estimation

Accurate covariance matrices for $\xi(r)$ measurements are critical for cosmological parameter inference. The current state of the art for DESI uses the RascalC code [18, 19], which computes a semi-analytical covariance in the Gaussian approximation (using the measured ξ and the random catalog to encode the survey geometry) and approximates non-Gaussian contributions by rescaling the shot-noise terms with a single parameter α_{SN}, calibrated from jackknife resampling or a small set of mocks.

The kNN framework interacts with this covariance pipeline at three levels:

(i) Drop-in compatibility. Since the kNN estimator gives the same $\xi(r)$ as LS (Section 3.3), any existing covariance matrix is directly applicable. No changes to the downstream pipeline are needed.

(ii) Dilution-ladder variance. The multiple subsamples at each dilution level provide an empirical covariance estimate, analogous to jackknife but with natural scale separation (Section 5.3). This can serve as an independent cross-check on mock-based or semi-analytical covariances.

(iii) Scale-dependent α_{SN} from kNN residuals. As developed in Section 7, the non-Poissonian residuals $\Delta_k(r_0)$ provide a direct measurement of the excess variance $\delta\sigma_N^2(V)$ at each scale, replacing the single-parameter α_{SN} with a data-driven, scale-dependent function. This is the most novel contribution of the kNN approach to the covariance problem: it upgrades the least constrained ingredient of the RascalC pipeline at no additional computational cost.

11.4 Computational cost

A fair comparison with current survey pipelines must account for the optimizations already in use. The Euclid 2PCF pipeline [34] uses tree-accelerated pair counting (kd-tree, octree, or linked-list methods) with OpenMP parallelization, split random catalogs to reduce the RR cost, and the Linear Construction method [35] to speed up mock-based covariance estimation. We compare the asymptotic scaling, but caution that wall-clock performance depends on constants, cache behavior, and parallelization efficiency that can only be determined by benchmarking (Section 10).

Operation	Tree-accelerated LS	kNN ladder
DD	$O(N_d \cdot N_{\text{shell}})$	$O(N_d \log N_d)$ per level
DR	$O(N_R \cdot N_{\text{shell}})$	$O(N_R \log N_d)$ per level
RR (uniform, unweighted)	$O(N_R^2)$ or split	Analytic (Erlang)
RR (weighted survey)	$O(N_R^2)$ or split	$O(N_R \log N_R)$
Mock-based covariance	$N_{\text{mock}}\times$ above	$N_{\text{mock}}\times$ above
Binning	Required (pre-chosen)	Not needed
Non-Poissonian diagnostics	Not available	Free (from same query)

Here $N_{\rm shell}$ is the mean number of objects in a radial shell at the separation of interest. For tree-accelerated pair counting, the effective $N_{\rm shell}$ depends on the tree pruning efficiency and is typically much smaller than the naive shell count, particularly at small separations where the tree prunes aggressively. At large separations ($r \gtrsim 100\,h^{-1}\,\mathrm{Mpc}$), $N_{\rm shell}$ grows as r^2 and tree acceleration provides diminishing returns; this is exactly where the kNN dilution ladder is most advantageous.

The kNN approach does not eliminate the need for mock-based covariance estimation. Mock catalogs are essential for validating the pipeline, characterizing systematic errors, and computing the model-dependent covariance at different points in parameter space. Whatever speedup the kNN method provides per catalog applies equally to the mock runs. The dilution-ladder variance (Section 5.3) is a complementary, data-driven diagnostic that provides an independent cross-check on the mock-based covariance but does not replace it.

Definitive wall-clock benchmarks at survey scale ($N_d \sim 2 \times 10^7$, $N_R \sim 10^9$) comparing bosque-based kNN queries to Corrfunc and the Euclid 2PCF code are deferred to the companion code paper.

11.5 Summary

We have shown that the k-Nearest Neighbor CDF framework provides an estimator for the two-point correlation function $\xi(r)$ that is:

- Mathematically equivalent to the Landy–Szalay estimator when the summed pair-count density is used with sufficient $k_{\rm max}$.
- Computationally faster by orders of magnitude, scaling as $O(N_R \log N_d)$ rather than $O(N^2)$.
- Naturally continuous in r, requiring no choice of radial binning.
- Compatible with survey geometry and per-object weights through the value-weighted kNN-CDF formalism.
- Extendable to large scales via the dilution ladder, with built-in overlap consistency checks and empirical variance estimation.
- Equivalent in statistical precision to pair counting, with the same cosmic variance floor and the same shot noise.

Beyond estimating $\xi(r)$ itself, the kNN framework provides—at no additional computational cost—a scale-dependent diagnostic of non-Poissonian structure in the galaxy point process (Section 7). The residual between the measured kNN-CDFs and the Poisson$|\xi$ prediction yields the excess factorial moments at each scale, directly measuring the non-Gaussian and non-Poissonian contributions that enter the covariance matrix. This enables a data-driven, scale-dependent replacement for the single-parameter shot-noise rescaling used in current semi-analytical covariance estimation (RascalC), potentially improving covariance accuracy for both BAO and full-shape analyses.

The method is immediately applicable to any survey that currently uses pair counting for $\xi(r)$ estimation. It requires only a standard KD-tree implementation and the survey random catalog. We encourage the LSS community to validate the approach on existing mock catalogs and adopt it where the computational savings are significant.

Code implementing the kNN ladder estimator, including survey weight support, the dilution scheme, and the non-Poissonian diagnostic, will be made publicly available.

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